

Mohammad Rayyan

Python & Web Developer · AI-Assisted Development

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SUMMARY

BCA student at Christ University, Bangalore, who builds fast, dependable software in Python and for the web. Works AI-first: uses agentic coding tools and modern LLMs every day to plan, build, review, and debug, and picks up new AI models and developer tools soon after release. Pairs that speed with solid engineering practice, including performance profiling, automated testing, and clean, well-documented code.

PROJECTS

Caramel · Offline voice-controlled Linux desktop agent — Nov 2026 – Present

- Built a fully offline voice assistant (no cloud APIs) using Silero VAD and Vosk/Kaldi speech recognition, with a 3-state engine (dormant → listening → executing).
- Designed a deterministic prefix-tree (trie) command engine with ~0.5 µs lookups, handling 53 voice commands with no collisions; also added dynamic Google/YouTube search.
- Kept the footprint small: ~0.47 ms ASR inference per 32 ms audio frame, under 2% CPU while idle, and 15–85 MB RAM.
- Shipped a one-command installer that detects installed apps, a systemd user service, a global *caramel* CLI, and 8 automated test suites.

`Python · Vosk · Silero VAD (ONNX) · Bash · systemd · pytest`

MawBooks · E-book publishing & reading platform (in progress) — Jul 2026 – Present

- Building a platform for independent authors to publish e-books and for readers to find them, with a PHP backend and session-based login.
- Designed a custom responsive UI with its own typography and branding; planned architecture adds a React front end, a MySQL/PDO data layer, and role-based admin moderation.

`PHP 8 · JavaScript · CSS3 · MySQL · Apache`

Sudoku Solver · Algorithmic puzzle solver — Sep 2026

- Wrote a recursive backtracking solver for 9×9 Sudoku that checks rows, columns, and 3×3 boxes at each step.

`Python · Recursion · Backtracking`

EDUCATION

Bachelor of Computer Applications (BCA) — 2025 – 2028 (expected)

Christ University, Central Campus, Bangalore · Currently 2nd year (2 BCA A) · Reg. No. 2641528

SKILLS

Languages	Python, PHP, JavaScript, HTML/CSS, Bash
AI & Tooling	Agentic IDEs (Antigravity), Gemini, ChatGPT, Claude, prompt engineering, Git & GitHub
Systems & Data	Linux, systemd, MySQL, Apache
Concepts	Speech recognition & VAD, data structures & algorithms, automated testing, performance profiling

INTERESTS

Machine automation · Cryptocurrency · Animation